# AI-Powered SOC

# Alert Analysis System

*AI-driven Security Monitoring & Incident Response System*

Lopamudra | Software Test Engineer | AI & Cybersecurity Enthusiast

# Problem Statement — Traditional SOC Challenges

## Alert Fatigue

- SOC analysts face 1000s of alerts daily
- Over 80% are false positives
- Critical threats get buried in noise

## Manual Log Analysis

- Logs from dozens of sources
- Hours spent on manual correlation
- Error-prone and inconsistent

## Slow Incident Response

- Mean time to detect: hours to days
- No automated escalation path
- Business impact keeps growing

**80%+ alerts are false positives**

**11min avg threat detection delay**

**$4.9M avg cost per breach (IBM 2023)**

# Solution Overview — AI-Powered Automation

**Automate Detection · Prioritize Threats · Generate Actionable Reports — All with AI**

### Ingest Logs

Multi-source ingestion

### Detect Threats

Rule + ML-based

### AI Classify

LLM-powered triage

### Auto Report

Structured outputs

*Key Benefit: From hours of manual analysis to seconds of AI-powered insight*

# System Architecture

End-to-End AI-Powered SOC Pipeline

## LOG SOURCES (INPUT)

- System Logs (/var/log)
- Network Logs (syslog)
- Wireshark PCAP

## INGESTION LAYER (PROCESS)

- Log Parser & Normaliser
- Event Queue (async)
- Schema Validator

## DETECTION ENGINE (ANALYSE)

- Snort Rule Matching
- Anomaly Detection
- Threshold Triggers

## AI ANALYSIS (AI/LLM)

- LLM Classification
- Root Cause Engine
- Severity Scoring

## OUTPUT LAYER (OUTPUT)

- Alert Management
- Incident Report (PDF)
- Dashboard Feed

RAW EVENTS → NORMALISED → FLAGGED → CLASSIFIED → REPORTED ✓

# Data Flow Diagram — Step-by-Step Pipeline

| 01 | 02 | 03 | 04 | 05 | 06 |
|---|---|---|---|---|---|
| **LOGS** | **PARSING** | **DETECTION** | **AI ANALYSIS** | **ALERT** | **REPORT** |
| Raw Events Collected | Normalise & Structure | Rules & Anomaly Match | LLM Classification | Severity-based Triage | Automated Incident Doc |
| syslog, /var/log, pcap | regex, JSON struct. | Snort rules, ML models | GPT-4 / Claude API | CRITICAL / HIGH / MED | PDF + JSON output |

# Key Features

## Log Ingestion

- Multi-source: syslog, network, Wireshark
- Real-time stream processing
- Structured normalisation pipeline

## Threat Detection

- Signature-based rule matching
- Behavioural anomaly detection
- Configurable thresholds per environment

## AI Classification

- LLM-powered alert triage
- Severity scoring (CRITICAL→LOW)
- Natural language root cause output

## Automated Reporting

- Structured JSON + PDF reports
- Timeline reconstruction
- Analyst-ready incident summaries

# AI Integration — Where Intelligence Is Applied

**Pattern Recognition**

Learns from historical events

**Alert Prioritisation**

CRITICAL / HIGH / MED / LOW

**AI / LLM Core**

**Root Cause Analysis**

Natural language explanation

**Report Generation**

Auto-written incident summaries

# Tech Stack

### Python 3.x

Core backend, parsing, orchestration

### Wireshark / tshark

Network packet capture & analysis

### Linux (Ubuntu)

System log sources, scripting env

### LLM API

GPT-4 / Claude for AI classification

### Snort / Rules

Signature-based threat detection

### JSON / PDF

Structured output and reporting format

# Real-World Mapping — SOC Workflow Alignment

| TRADITIONAL SOC | | THIS SYSTEM |
|---|---|---|
| Analyst Monitors | → | **Automated Log Ingestion** |
| Alert Triggered | → | **Detection Engine Fires** |
| Investigation Begins | → | **AI Root Cause Analysis** |
| Report Written | → | **Auto Report Generated** |

*Every manual step in the traditional SOC is replaced or accelerated by intelligent automation*

# Sample Use Case — Brute Force Attack Detection

**Scenario: Multiple failed SSH login attempts detected from 192.168.1.105**

## 1 Log Ingestion

- /var/log/auth.log streamed into system
- 50 failed SSH login events in 60 seconds

## 2 Detection

- Threshold rule triggers: >10 fails/min
- Source IP flagged for investigation

## 3 AI Analysis

- LLM classifies: Brute Force Attack
- Severity: CRITICAL | Confidence: 96%

## 4 Auto Report

- Incident report generated in 2 seconds
- Recommendation: Block IP, review accounts

**Result: Threat detected and reported in < 3 seconds | Zero analyst intervention required | Full audit trail preserved**

# Future Enhancements

## SIEM Integration

*Splunk / IBM QRadar*

- Bi-directional alert sync
- Centralised log management
- Enterprise-grade scalability

## Real-time Dashboard

*Live threat visibility*

- WebSocket-powered live feed
- Geo-IP threat mapping
- Executive KPI summaries

## Cloud Scaling

*AWS / Azure deployment*

- Kubernetes orchestration
- Elastic log ingestion
- Multi-region redundancy

*Current build is a working MVP · Roadmap targets enterprise-grade deployment*

# "Bridging QA, AI, and Cybersecurity to build intelligent monitoring systems"

- ✓ Demonstrated AI + Security integration at system level
- ✓ Full pipeline: ingestion → detection → classification → reporting
- ✓ QA mindset applied to build reliable, testable, observable systems

Lopamudra | Software Test Engineer | AI & Cybersecurity Enthusiast

Open to AI · Cybersecurity · QA Automation roles